Round 1 Student Challenge Brief

Manufacturing Manual Assistant using Python Full Stack and RAG

Your challenge

A manufacturing organization stores equipment manuals in separate PDF files. Engineers need a faster way to find instructions for a selected machine. Build a web application that retrieves relevant passages and generates answers grounded in those passages, with verifiable document and page references.

Round details

Item	Requirement
Team size	3 or 4 students
Development	Day 1, 12:00 PM to 6:30 PM; 6.5 hours
Submission deadline	6:30 PM on Day 1
Evaluation	Day 1, 7:00 PM to 10:00 PM

Expected user workflow

- Administrator: log in, register machines, and upload manuals with model and revision details.
- Engineer: log in, select a machine, ask a maintenance question, read the answer, and open the cited PDF page.
- Supervisor: review question history and identify unanswered queries. A simple read-only view is sufficient.

Independent work

Your team chooses its design, task allocation, and implementation sequence. Supervision is limited to clarifying requirements and addressing common environment issues. Every member must explain their contribution. Follow the college's announced policy on AI coding assistants and reused code.

Functional Requirements and Reference Files

- Build a browser interface, Django backend, Django REST Framework APIs, and PostgreSQL or MySQL persistence. Use HTML, CSS, JavaScript, and Bootstrap for the interface.
- Enforce roles on the server. Only Administrators upload manuals. Engineers use the assistant and access their saved history; Supervisors can review the question log.
- Extract text from the supplied PDFs using PyMuPDF. Preserve machine ID, document ID, revision, and original PDF page number with every searchable passage.
- Use embeddings, a vector store, and an LLM for RAG. Restrict retrieval to the selected machine before generating the answer. Both HP-200 manuals are valid complementary sources.
- Show a supporting passage, document name or ID, revision, PDF page number, and source link. Cite each source needed for a multi-document answer. Do not invent citations.
- If evidence is missing, state that the available manuals do not contain enough information. Do not answer using another machine's manual or general model knowledge.
- Persist question, answer, selected machine, timestamp, source references, and an explicit answered or insufficient-evidence status.

- Include two automated checks: Engineers cannot upload manuals, and retrieval excludes passages belonging to another machine.

Files supplied

PDF file	Machine and purpose
HP200_Operations_Rev1_0.pdf	M-HP200 | HP-200 | Operating overview, limits, alarms
HP200_Maintenance_Rev1_0.pdf	M-HP200 | HP-200 | Inspection schedule and maintenance troubleshooting
AC90_Operations_Maintenance_Rev1_0.pdf	M-AC90 | AC-90 | Operating overview, limits, alarms, and maintenance

Use machine_manual_index.csv as registration metadata. Index only the three PDFs in Manuals as the answer corpus. Do not index this challenge brief, START_HERE.txt, or the metadata CSV as maintenance guidance. Page 1 of each PDF counts as PDF page 1; the printed page numbers match this convention.

You (the student team) will use your embedding model and LLM access separately. Submission and Demonstration

Demonstration checklist

- Ask a supported question and open the cited page.
- Ask a question requiring information from the two HP-200 manuals.
- Ask a question whose evidence exists only for the other machine; show that the selected machine remains isolated.
- Ask a question that the supplied manuals cannot answer.
- Attempt an upload as an Engineer and show denial; reopen stored history as the appropriate user.

Practice questions

These are unscored examples; judges will also use unseen questions. Select the machine before asking: “What does E17 mean?”, “What is the normal operating pressure range?”, “How often is the filter checked?”, and “What should be recorded before escalation?” Your response must follow the selected machine’s documents.

For example: Ask the application: How often should the AC-90 AF-10 filter indicator be checked, and when is the planned filter replacement due?